05/07/2022

CODE-A

FINAL TEST SERIES for NEET-2022

MM : 720

Test - 10

Time : 3 Hrs. 20 Mins.

Answers

1. (1)	41. (3)	81. (3)	121. (2)	161. (4)
2. (2)	42. (1)	82. (4)	122. (4)	162. (1)
3. (3)	43. (1)	83. (4)	123. (1)	163. (3)
4. (3)	44. (2)	84. (1)	124. (1)	164. (4)
5. (4)	45. (2)	85. (3)	125. (3)	165. (4)
6. (4)	46. (4)	86. (3)	126. (3)	166. (2)
7. (2)	47. (1)	87. (3)	127. (3)	167. (2)
8. (4)	48. (3)	88. (1)	128. (1)	168. (3)
9. (1)	49. (3)	89. (1)	129. (2)	169. (2)
10. (1)	50. (2)	90. (3)	130. (3)	170. (3)
11. (1)	51. (3)	91. (4)	131. (1)	171. (2)
12. (2)	52. (2)	92. (4)	132. (2)	172. (3)
13. (1)	53. (3)	93. (3)	133. (3)	173. (2)
14. (2)	54. (2)	94. (4)	134. (1)	174. (1)
15. (2)	55. (4)	95. (3)	135. (4)	175. (1)
16. (4)	56. (2)	96. (2)	136. (2)	176. (4)
17. (1)	57. (4)	97. (3)	137. (4)	177. (3)
18. (4)	58. (3)	98. (3)	138. (3)	178. (2)
19. (2)	59. (1)	99. (4)	139. (4)	179. (4)
20. (2)	60. (1)	100. (4)	140. (4)	180. (2)
21. (2)	61. (2)	101. (2)	141. (3)	181. (1)
22. (3)	62. (3)	102. (2)	142. (1)	182. (1)
23. (1)	63. (2)	103. (3)	143. (2)	183. (1)
24. (4)	64. (4)	104. (2)	144. (3)	184. (1)
25. (1)	65. (2)	105. (4)	145. (4)	185. (4)
26. (1)	66. (1)	106. (4)	146. (2)	186. (3)
27. (3)	67. (2)	107. (3)	147. (2)	187. (4)
28. (1)	68. (2)	108. (3)	148. (3)	188. (4)
29. (2)	69. (4)	109. (1)	149. (2)	189. (1)
30. (4)	70. (2)	110. (2)	150. (2)	190. (3)
31. (4)	71. (3)	111. (3)	151. (1)	191. (4)
32. (3)	72. (3)	112. (3)	152. (2)	192. (1)
33. (3)	73. (1)	113. (3)	153. (2)	193. (2)
34. (1)	74. (1)	114. (3)	154. (3)	194. (3)
35. (4)	75. (4)	115. (3)	155. (2)	195. (2)
36. (1)	76. (1)	116. (1)	156. (2)	196. (4)
37. (1)	77. (4)	117. (4)	157. (4)	197. (3)
38. (4)	78. (2)	118. (2)	158. (3)	198. (1)
39. (3)	79. (3)	119. (2)	159. (4)	199. (2)
40. (1)	80. (4)	120. (3)	160. (3)	200. (1)

FINAL TEST SERIES for NEET-2022

MM : 720 **Test - 10** Time : 3 Hrs. 20 Mins.

Hints and Solutions

PHYSICS

SECTION - A

1. Answer (1)

$u_x = 20\cos 37°$

$= 20 \times \frac{4}{5} = 16$ m/s

$u_y = 20\sin 37°$

$= 20 \times \frac{3}{5} = 12$ m/s

$v_x = u_x$

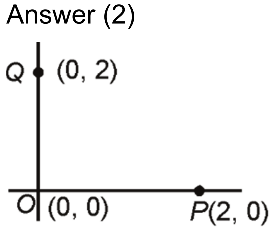
$= 16$ m/s

$v_y = u_y - gt$

$= 12 - 10 \times \frac{1}{2} = 7$ m/s

$\vec{v} = \left(16\hat{i} + 7\hat{j}\right)$ m/s

2. Answer (2)



$$\overrightarrow{r_{com}} = \frac{m_1\vec{r_1} + m_2\vec{r_2} + m_3\vec{r_3}}{m_1 + m_2 + m_3}$$

$$= \frac{2 \times 0 + 2 \times 2\hat{i} + 2 \times 2\hat{j}}{6}$$

$$= \left(\frac{2}{3}\hat{i} + \frac{2}{3}\hat{j}\right) \text{ m}$$

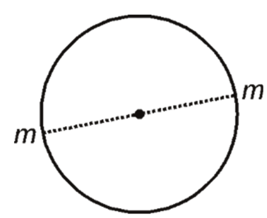
3. Answer (3)

$$\text{Rotantional kinetic energy} = \left(\frac{\frac{1}{2}I\omega^2}{\frac{1}{2}I\omega^2 + \frac{1}{2}mv^2}\right)mgh_0$$

$$= \left(\frac{\frac{1}{2}\cdot\frac{2}{3}mR^2\,\frac{v^2}{R^2}}{\frac{1}{2}\cdot\frac{2}{3}mR^2\,\frac{v^2}{R^2} + \frac{1}{2}mv^2}\right)mgh_0$$

$$= \frac{2}{5} \times mgh_0$$

4. Answer (3)



$$F_g = \frac{mv^2}{R}$$

$$\frac{Gmm}{(4r)^2} = \frac{mv^2}{2r}$$

$$v = \sqrt{\frac{Gm}{8r}}$$

$$v = \frac{1}{2}\sqrt{\frac{Gm}{2r}}$$

5. Answer (4)

$$T = \frac{2\pi}{\omega}$$

$\omega = 300$

$$T = \frac{\pi}{150}$$

6. Answer (4)

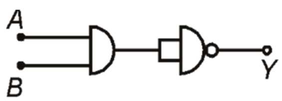
In UCM velocity of particle is constant in magnitude but its direction changes continuously.

Hence, $|\Delta\vec{P}| \neq 0$

7. Answer (2)

$n_f = 2$, $n_i = 4$ transition photon will be having least wavelength.

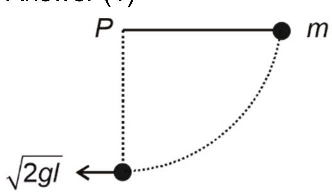
8. Answer (4)



$$Y = \overline{(A \cdot B) \cdot (A \cdot B)}$$

$$Y = \overline{A \cdot B}$$

9. Answer (1)



By conservation of energy

$$mgl = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gl}$$

$$|\vec{L}| = mvl$$

$$= ml\sqrt{2gl}$$

10. Answer (1)

$$V_A = \frac{KQ}{a} - \frac{KQ}{\sqrt{5}a} = \frac{KQ}{a}\left[1 - \frac{1}{\sqrt{5}}\right]$$

$$V_B = -\frac{KQ}{a} + \frac{KQ}{3a} = \frac{-2KQ}{3a}$$

$$V_A - V_B = \frac{KQ}{a} - \frac{KQ}{\sqrt{5}\,a} + \frac{2\,KQ}{3\,a}$$

$$= \frac{KQ}{a}\left[\frac{5}{3} - \frac{1}{\sqrt{5}}\right]$$

11. Answer (1)

$$f_A = f + \frac{f}{50}$$

$$f_B = f - \frac{f}{50}$$

$$f_A - f_B = \frac{f}{50} + \frac{f}{50}$$

$$\frac{f}{25} = 8$$

$f = 200$ Hz

$$f_B = 200 - \frac{200}{50}$$

$= 196$ Hz

12. Answer (2)

$V_{IG} = 10$ m/s

$$\overrightarrow{V_{I/M}} = -\overrightarrow{V_{O/M}}$$

$$\left|\overrightarrow{V_{IM}}\right| = 10 \text{ m/s}$$

13. Answer (1)

$$\sin i_c = \frac{1}{\mu}$$

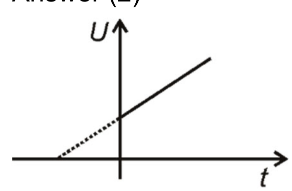
$$\sin 37° = \frac{1}{\mu}$$

$$\mu = \frac{5}{3}$$

$\tan i_p = \mu$

$$i_p = \tan^{-1}\left(\frac{5}{3}\right)$$

14. Answer (2)



$$U = \frac{1}{2}nfRT$$

$T = (t + 273)$

$$U = \frac{1}{2}nfR\,(t + 237)$$

$y = mx + C$

15. Answer (2)

$$\phi = \frac{12400 \text{ eV}}{4000 \text{ Å}} = 3.1 \text{ eV}$$

$\phi_0 = \phi - \text{K.E.}$

$= (3.1 - 1.68)$ eV

$= 1.42$ eV

16. Answer (4)

$$\lambda = \frac{h}{mv}$$

$$v_n \propto \frac{1}{n}$$

17. Answer (1)

Alloys, due to their high resistivity, are used to make standard resistance coils.

18. Answer (4)

As force will act all the time in cyclotron hence, particle undergoes acceleration all time and work is done by only electric field. Hence, gain in kinetic energy occurs only between the dees.

19. Answer (2)

$$E = \frac{2KP}{r^3}$$

20. Answer (2)

$P \propto \frac{1}{r} \Rightarrow$ 40 W bulb has larger resistance. Thus, it will fuse first if same current flows through both the resistances.

21. Answer (2)

$$B = \frac{\mu_0}{4\pi}\frac{\alpha i}{R} = \frac{3\mu_0 i}{8R}$$

22. Answer (3)

$${}_{92}U^{237} \xrightarrow[\text{emissions}]{7\alpha\text{-particle}} {}_{78}Pt^{209} \xrightarrow[\text{emissions}]{5\beta\text{-particle}} {}_{83}Bi^{209}$$

23. Answer (1)

In Lyman series H_β line correspond to transition from state $n_i = 3$ to state $n_f = 1$.

$\Rightarrow$ Energy emitted $E = E_3 - E_1$

$$= \frac{-13.6}{3^2} - \frac{-13.6}{1^2}$$

= (– 1.51 + 13.6) eV

= 12.09 eV

24. Answer (4)

As charge will not change and capacity will decrease.

$U = \frac{Q^2}{2C}$, energy will increase.

25. Answer (1)

Angle of repose $\theta = \tan^{-1}\mu = \tan^{-1}(0.7)$

$\Rightarrow$ Angle of repose > 30°, so frictional force = $mg \sin 30° = 25$ N

26. Answer (1)

The unit of $\frac{QR}{L}$ will be those of current.

27. Answer (3)

Amplitude $(A) = \dfrac{30}{2 + (x - 20t)^2}$

Denominator should be minimum

$\therefore$ for $x - 20t = 0$

$$A = \frac{30}{2} = 15 \text{ m}$$

28. Answer (1)

$$\frac{1}{f_l} = (\mu - 1)\left(\frac{2}{R}\right)$$

$$\frac{1}{20} = \frac{0.5 \times 2}{R}$$

$R = 20$ cm

$$\frac{1}{f'} = (1.5 - 1)\left(\frac{1}{-20} - \frac{1}{20}\right)$$

$$\frac{1}{f'} = (0.5) \times \left(\frac{-2}{20}\right) = \frac{-1}{20}$$

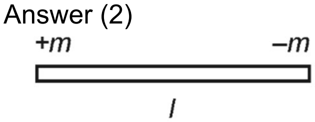
Power of arrangement (P) = $\frac{1}{f} + \frac{1}{f'} + \frac{1}{f}$

$$= \frac{1}{20} + \left(\frac{-1}{20}\right) + \frac{1}{20}$$

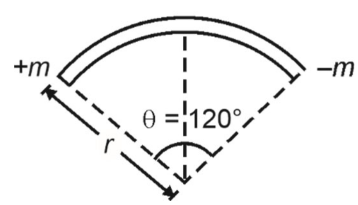
$$= \frac{1}{20}$$

$P = 5$ D

29. Answer (2)



$M = ml$



$$r\left(\frac{2\pi}{3}\right) = l$$

$$r = \frac{3l}{2\pi}$$

Now, $M' = m \times r \sin 60 \times 2$

$$= 2m \times \frac{3l}{2\pi} \times \frac{\sqrt{3}}{2}$$

$$M' = \frac{3\sqrt{3}\,M}{2\pi}$$

30. Answer (4)

Electric Field (E) = $\dfrac{\Delta V}{L}$

Energy = $e(\Delta V) = 2 \times 1.6 \times 10^{-19}$ J

$\Delta V = 2$ V

Electric field (E) = $\dfrac{\Delta V}{L}$

$$= \frac{2}{4 \times 10^{-8}}$$

$= 0.5 \times 10^8$

$= 5 \times 10^7$ V/m

31. Answer (4)

$$\left(\frac{T}{V}\right)^2 V = \text{constant}$$

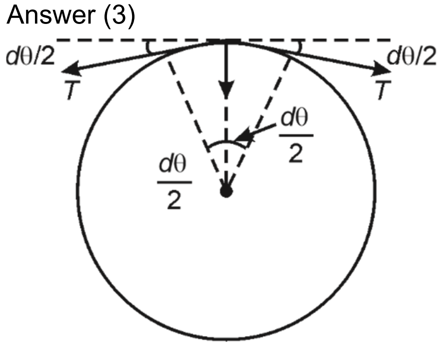
$$\frac{T^2}{V} = \text{constant}$$

$$\frac{T_1^2}{V_1} = \frac{T_2^2}{V_2}$$

$$\frac{T_0^2}{V_0} = \frac{T_2^2}{3V_0}$$

$$T_2 = \sqrt{3}T_0$$

32. Answer (3)



$$2T\sin\left(\frac{d\theta}{2}\right) = dmR\frac{v^2}{R^2} = \frac{m}{2\pi R}\frac{Rd\theta \times Rv^2}{R^2}$$

$$\sin\left(\frac{d\theta}{2}\right) \simeq \frac{d\theta}{2}$$

$$T = \frac{mv^2}{2\pi R}$$

33. Answer (3)

P = constant

$F.v = K$

$$m.v.v\frac{dv}{ds} = K$$

$$\int_0^v mv^2 dv = \int_0^s K ds$$

$$\frac{mv^3}{3} = Ks$$

$$v^3 = \left(\frac{3K}{m}\right)s$$

$$\frac{ds}{dt} = \left(\frac{3K}{m}\right)^{1/3} s^{1/3}$$

$$\int_0^s s^{-1/3} ds = \left(\frac{3K}{m}\right)^{1/3} \int_0^t dt$$

$$\frac{s^{-1/3+1}}{-1/3+1} = \left(\frac{3K}{m}\right)^{1/3} t$$

$$\frac{3}{2}s^{2/3} = \left(\frac{3K}{m}\right)^{1/3} t$$

$$s^{2/3} = \frac{2}{3}\left(\frac{3K}{m}\right)^{1/3} t$$

$s \propto t^{3/2}$

34. Answer (1)

Magnetic moment = IA

$$= I\left(\frac{\pi R^2}{2} + \pi\left(\frac{4R^2}{2}\right)\right)$$

$$= \frac{5\pi R^2 I}{2}$$

35. Answer (4)

$$I = \frac{20}{5} = 4 \text{ A}$$

Both resistances are equal

$\therefore$ current will be same i.e. 2A (each)

SECTION - B

36. Answer (1)

$$\frac{4}{3}\pi R^3 = 125 \times \frac{4}{3}\pi r^3$$

$$r = \frac{R}{5}$$

$E = T \times 4\pi R^2$

$$E_1 = T \times 125 \times 4\pi\left(\frac{R}{5}\right)^2$$

$= 5 \times E$

37. Answer (1)

$$V_{\text{rms}} = \sqrt{\frac{\int_0^{T/2} V^2\, dt}{\int_0^{T/2} dt}} = \frac{V_0}{\sqrt{3}}$$

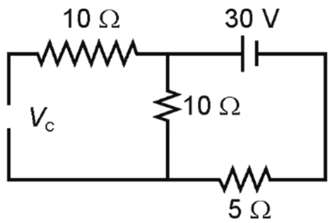
38. Answer (4)

$V_{\text{immersed}}\ \rho_w g = V\rho_w g$

$$\frac{V_{\text{immersed}}}{V} = \frac{\rho}{\rho_w} = x_0$$

Fraction immersed in liquid is independent of acceleration of vessel.

39. Answer (3)



$$I = \frac{30}{15}$$

$I = 2$ A

$V_c = 10 \times I$

$= 10 \times 2$

$= 20$ V

40. Answer (1)

$n^2 = n_e n_h$

$$n_e = \frac{\left(1.5\times10^{16}\right)^2}{4.5\times10^{22}}$$

$n_e = 5 \times 10^9\ \text{m}^{-3}$

41. Answer (3)

As $R^3 \propto A$

$\therefore\ R \propto A^{1/3}$

42. Answer (1)

$$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB} = \frac{\sqrt{2mqV}}{qB}$$

$$r \propto \sqrt{\frac{m}{q}}$$

43. Answer (1)

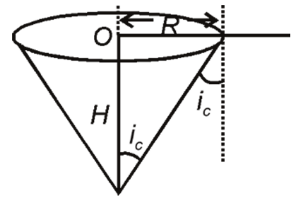
$$W = k\int_a^{2a} x^2 dx$$

$$W = k\left[\frac{x^3}{3}\right]_a^{2a}$$

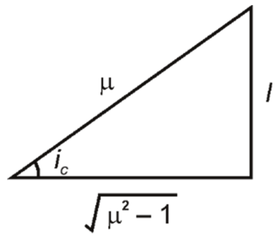
$$= \frac{k}{3}\left[8a^3 - a^3\right]$$

$$= \frac{k}{3}\times 7a^3$$

44. Answer (2)



$$\sin i_c = \frac{1}{\mu}$$



$$\tan i_c = \frac{1}{\sqrt{\mu^2 - 1}}$$

$$\frac{R}{H} = \frac{1}{\sqrt{\mu^2 - 1}}$$

$$R = \frac{H}{\sqrt{\mu^2 - 1}}$$

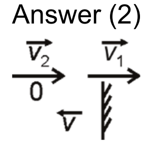
$$\frac{\pi H^2}{\mu^2 - 1} = (2H)^2$$

$$\frac{\pi}{4} = \mu^2 - 1$$

$$\mu^2 - 1 = \frac{\pi}{4}$$

$$\boxed{\mu = \sqrt{\frac{\pi}{4} + 1}}$$

45. Answer (2)



$\vec{v}_2 - \vec{v}_1 = \vec{u}_1 - \vec{u}_2$

$v_2 - 4 = 4 - 5$

$v_2 = 3$ m/s

46. Answer (4)

$$\frac{1}{R} = \frac{1}{3} + \frac{1}{16} + \frac{1}{8}$$

$$\frac{1}{R} = \frac{16 + 3 + 6}{48} = \frac{25}{48} \Rightarrow R = \frac{48}{25}$$

$$V_{PQ} = 1.5 \times \frac{48}{25} = \frac{14.4}{5} = 2.88\ \text{V}$$

47. Answer (1)

$\vec{L} = \vec{r} \times \vec{P}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 3 & 4 & -2 \end{vmatrix} = \hat{i}\,(-4+4) - \hat{j}\,(-2+3) + \hat{k}\,(4-6)$$

$$= 0\hat{i} - \hat{j} - 2\hat{k}$$

Angular momentum is perpendicular to *x*-axis.

48. Answer (3)

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{1}{\lambda} = \frac{10^5}{10^{-2}} \times (1)^2\left(\frac{1}{2^2} - \frac{1}{4^2}\right)$$

$$\nu = \frac{c}{\lambda} = 3\times10^8 \times 10^7 \times \left(\frac{1}{4} - \frac{1}{16}\right)$$

$$= 3\times10^{15} \times \frac{3}{16}$$

$$\nu = \frac{9}{16}\times10^{15}\text{ Hz}$$

49. Answer (3)

$N = N_0e^{-\lambda t}$

Activity at time t_1 is A_1

$$A_1 = \left|\frac{dN}{dt}\right| = N_0\lambda e^{-\lambda t_1}$$

Activity at time t_2 is A_2

$$A_2 = N_0\lambda e^{-\lambda t_2}$$

Number of nuclei decayed

$$= N_1 - N_2 = N_0e^{-\lambda t_1} - N_0e^{-\lambda t_2}$$

$$= N_0\left(e^{-\lambda t_1} - e^{-\lambda t_2}\right)$$

$$= \frac{A_1 - A_2}{\lambda}$$

50. Answer (2)

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.98}{1-0.98} = \frac{0.98}{0.02}$$

$\beta = 49$

CHEMISTRY

SECTION - A

51. Answer (3)

For n + l = 5, one 5s, three 4p and five 3d orbitals are possible.

52. Answer (2)

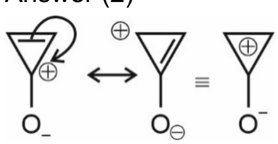
Generally ionization enthalpy increases from left to right in a period and decreases down the group.

53. Answer (3)

$\Delta H_{isomerisation} = (\Delta H_C)_R - (\Delta H_C)_P$

$= -1359.2 - (-1336.0) = -23.2$ kJ/mol

54. Answer (2)



To gain aromaticity, given molecule remains in ionic form.

55. Answer (4)

Isolated alkenes do not undergo birch reduction.

56. Answer (2)

Gold sol is negatively charged sol which is flocculated by positive ion having highest positive charge.

57. Answer (4)

Aromatic aldehydes do not respond to Fehling's solution test.

58. Answer (3)

$$CHCl_3 + NaOH \rightarrow \overset{\ominus}{C}Cl_3 \xrightarrow{-Cl^-} :CCl_2 \text{ (Reactive intermediate)}$$

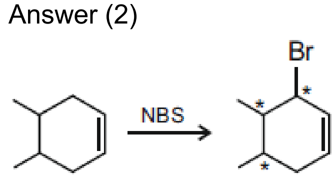
59. Answer (1)

Nylon 6 : $\left(\overset{O}{\overset{\|}{C}} - (CH_2)_5 - \overset{H}{\overset{|}{N}} \right)_n$

60. Answer (1)

Sugar having free –OH group at anomeric carbon is reducing in nature.

61. Answer (2)



62. Answer (3)

Vapour pressure order:

$Hg < H_2O < C_2H_5OH < C_6H_{14}$

63. Answer (2)

	N_2O_4	$\rightleftharpoons$ $2NO_2$
t = 0	1 mol	0 mol
t = $t_{eq.}$	0.4 mol	1.2 mol

$(M_{avg})_{eq.} = \dfrac{0.4\times92+1.2\times46}{1.6} = 57.5$

$(V.D.)_{eq.} = \dfrac{57.5}{2} = 28.75$

64. Answer (4)

CH_3COONa is a salt of weak acid and strong base whose pH = 7 + $\frac{1}{2}$ (pK_a + logC)

65. Answer (2)

$KMnO_4$ is used as self-indicator.

66. Answer (1)

	A(g) ⟶	B(g) +	C(g)
At t = 0	200	0	0
At t = 20 min	200 – p	p	p

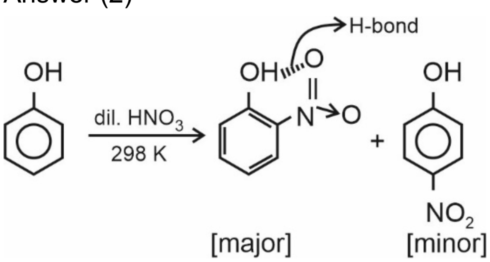
Given: 200 – p + p + p = 300

⇒ p = 100 mmHg

So, t = 20 minutes is the half-life period of the reaction

$\therefore \quad K = \dfrac{0.693}{t_{1/2}} = \dfrac{0.693}{20} \simeq 0.035\ min^{-1}$

67. Answer (2)



68. Answer (2)

$\overset{+}{N}R_3$ exerts -I effect

69. Answer (4)

HClO > $HClO_2$ > $HClO_3$ > $HClO_4$ [Oxidising power].

70. Answer (2)

$-\overset{\overset{O}{\|}}{C}-CH_3$ is a meta directing group.

71. Answer (3)

$K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-}$

1	0	0
0.9	0.4	0.1

i = 0.9 + 0.4 + 0.1 = 1.4

72. Answer (3)

$H_2SO_4 \rightarrow H^+ + HSO_4^-$

Anode : $2HSO_4^- \rightarrow H_2S_2O_8 + 2e^-$

Cathode : $2H^+ + 2e^- \rightarrow H_2$

73. Answer (1)

HNO_3 manufacture by Ostwald's process.

74. Answer (1)

More stable is the carbocation, greater is the rate of S_N1, benzyl carbocation formed in the reaction is most stable.

75. Answer (4)

- 2 g Fe atoms $= \frac{2}{56}$ mol atoms
- 2 g H_2O molecules = $\frac{2}{8}$ mole molecules = $\frac{6}{18}$ mole atoms
- 16 g CH_4 = 1 mole molecules = 5 mole atoms
- 10000 mL H_2O = $\frac{10000}{18} = 555.5$ mole molecules
  = 555.5 × 3 mole atoms

76. Answer (1)

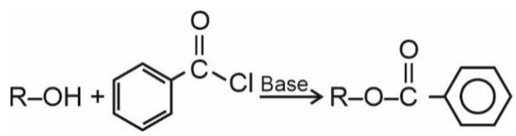
O_2 contains two unpaired electrons hence paramagnetic.

77. Answer (4)

H_2^{2-} has zero bond order.

78. Answer (2)

Schotten-Baumann reaction



79. Answer (3)

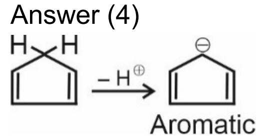
$C_6H_6 + C_6H_5CH_3 \rightarrow$ Ideal solution

n-heptane + n-hexane → Ideal solution

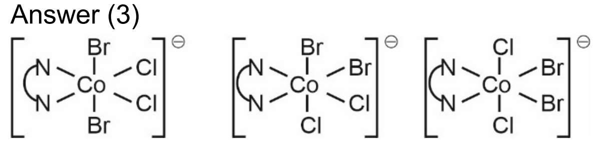
$C_2H_5Cl + C_2H_5Br \rightarrow$ Ideal solution

$H_2O + C_2H_5OH \rightarrow$ +ve deviation

80. Answer (4)



81. Answer (3)

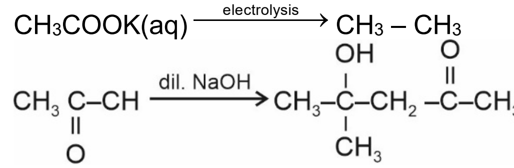


∴ 3 Geometrical isomers.

82. Answer (4)

- $NaHCO_3$ does not react with aldehyde.
- $NaHCO_3$ releases CO_2(g) with stronger acids then carbonic acid.

83. Answer (4)

- $CH_3COOK(aq) \xrightarrow{\text{electrolysis}} CH_3 - CH_3$
- 

- $CH_3MgBr \xrightarrow[\text{(ii) } H_3O^{\oplus}]{\text{(i) } CH_3CH_2CHO} CH_3-\underset{H}{\overset{OH}{C}}-CH_2-CH_3$
- $CH_3-CH_2-CH_2MgBr \xrightarrow{H_2O}$

$CH_3-CH_2-CH_3$

84. Answer (1)

Charge × distance for C–F bond is more than that of C–Br bond there tore dipole moment of CH_3I is greater than CH_3BR.

85. Answer (3)

Zinc with dilute HCl.

$Zn + 2H^{\oplus} \rightarrow Zn^{+2} + H_2(g)$

SECTION - B

86. Answer (3)

$$H_3BO_3 \xrightarrow{\Delta} \underset{[A]}{HBO_2} \xrightarrow{\Delta} \underset{[B]}{B_2O_3}$$

87. Answer (3)

Six large chloride ions cannot be accommodated around Si^{+4} due to limitation of its size.

88. Answer (1)

Bond strength order:

S – S > O – O > Se – Se > Te – Te

89. Answer (1)

Trigonal pyramidal $\Rightarrow XeO_3$

Square pyramidal $\Rightarrow XeOF_4$

Linear $\Rightarrow XeF_2$

Distorted octahedral $\Rightarrow XeF_6$

90. Answer (3)

$LiHCO_3$ is not obtained in the solid form, while all other alkali metals form solid hydrogen carbonates.

91. Answer (4)

$$Ca(OH)_2 + \underset{[A]}{Cl_2} \longrightarrow CaCl_2 + \underset{[B]}{Ca(OCl)_2} + H_2O$$

92. Answer (4)

Greenhouse gases:

[CH_4, H_2O, CFCs, O_3 and N_2O]

93. Answer (3)

$$X_{co} = \frac{n_{co}}{\text{Total mole}} = \frac{2}{2+1} = \frac{2}{3}$$

P_{co} = Total pressure × X_{co}

$$= 1 \times \frac{2}{3} = \frac{2}{3} \text{ atm} = 0.67 \text{ atm}$$

94. Answer (4)

$\because$ $2H_2 \rightarrow 4H + x$ kJ

$\therefore$ [H – H] bond energy is $\frac{x}{2}$kJ.

95. Answer (3)

$$= \frac{\frac{2 \times 4}{3}\pi r^3}{a^3} \qquad \because [4r = \sqrt{3}a]$$

$$= \frac{\frac{2 \times 4}{3}\pi r^3}{\left(\frac{4r}{\sqrt{3}}\right)^3} = \frac{\pi\sqrt{3}}{8}$$

96. Answer (2)

$CaO + SiO_2 \rightarrow CaSiO_3$

[flux] [Slag]

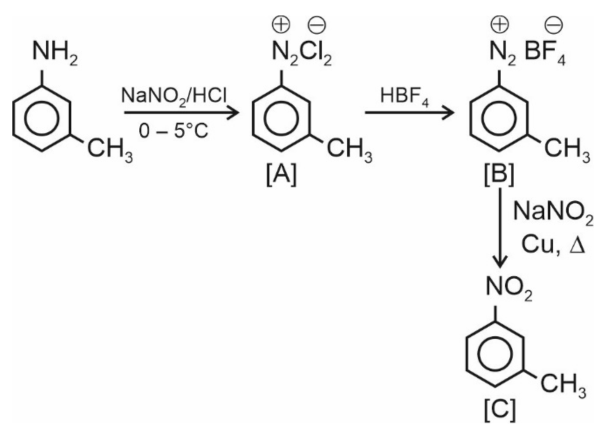
97. Answer (3)

In $[NiCl_4]^{-2}$ Ni is Ni^{+2} [$3d^8$] and Cl^- is a weak field ligand. Thus, it has 2 unpaired electrons,

⇅	⇅	⇅	↑	↑

$3d^8$. Hence, paramagnetic.

98. Answer (3)



99. Answer (4)

0.2% solution of phenol is used as an antiseptic.

100. Answer (4)

- Keratin and myosin are fibrous proteins.
- Insulin and albumins are globular proteins.

BOTANY

SECTION - A

101. Answer (2)

Poales is an order of wheat

102. Answer (2)

Fabaceae is a family of *Sesbania*, a kind of fodder.

103. Answer (3)

Protists are not chemoautotrophic.

104. Answer (2)

Chemosynthetic bacteria are chemoautotrophs. They use energy released by oxidation of certain inorganic substance for ATP production and play a great role in nutrient recycling.

105. Answer (4)

Methanogens produce methane.

106. Answer (4)

Six kingdom classification was given by Carl Woese based on sequence of 16S rRNA genes.

107. Answer (3)

Basidiospores and ascospores are sexual spores. Zoospores are endogenously produced.

108. Answer (3)

Pea has marginal placentation whereas tomato has axile placentation.

109. Answer (1)

All tissues outside the vascular cambium constitute the bark. Pith and secondary xylem do not constitute the bark.

110. Answer (2)

Ladybird and dragonflies are useful to get rid of aphids and mosquitoes. *i.e*. they are biocontrol agents.

111. Answer (3)

Megaspore mother cell is formed in the micropylar region of the nucellus.

112. Answer (3)

Population growing in a habitat with unlimited resources represent exponential growth with J-shaped curve, described by $\frac{dN}{dt} = rN$

113. Answer (3)

DDT concentration increases at successive trophic levels.

114. Answer (3)

Co-evolution is not a cause of biodiversity loss.

It is a process of reciprocal evolutionary change that occur in interacting species.

115. Answer (3)

Anticodon - UAC (for methionine) is present on tRNA.

116. Answer (1)

$$\left.\begin{array}{l} AABB \\ AAbb \\ aaBB \\ aabb \end{array}\right\} \text{Homozygous for both traits}$$

Total = 16

Possible outcomes = 4

$\therefore$ 4/16

117. Answer (4)

Meristem culture produces pathogen free clones of plant.

118. Answer (2)

On newly cooled volcanic lava that forms igneous rocks, primary succession can start.

119. Answer (2)

Euglena and *Paramoecium* have pellicle but lack cell wall.

120. Answer (3)

Cambium is absent in monocots.

121. Answer (2)

Bulliform cells are large, empty and colourless.

122. Answer (4)

Correct relation is $\psi_w = \psi_s + \psi_p$

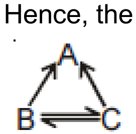
123. Answer (1)

For A = ψ_w = –9

For B = ψ_w = –5

For C = ψ_w = –5

Water moves from high to low water potential. Hence, the flow will be as follows:



124. Answer (1)

First CO_2 fixing enzyme in C_4 plants is present in cytoplasm of mesophyll cells.

125. Answer (3)

Glycolysis is common for both anaerobic and aerobic respiration.

126. Answer (3)

Chlorophyll and cytochromes are synthesized by succinyl CoA.

Acetyl CoA is raw material for gibberellins.

α-ketoglutaric acid is raw material for the synthesis of amino acids.

127. Answer (3)

Sporophytes of mosses are more elaborate than that of liverworts.

Pteridophytes have multicellular gametophyte and *Pinus* in monoecious.

128. Answer (1)

The basal body of cilia and flagella in eukaryotes has nine peripheral fibrils.

9 + 2 arrangement of microtubules is found in axoneme of eukaryotic cilia and flagella.

129. Answer (2)

Eutrophication is natural aging of a lake due to nutrient enrichment.

130. Answer (3)

Mitochondria	–	Oxidative phosphorylation
Chloroplast	–	Calvin cycle
Ribosomes	–	Protein synthesis

131. Answer (1)

Trichoderma polysporum produces cyclosporin A an immunosuppressive agent.

132. Answer (2)

C_4 plants avoid photorespiratory losses as well as they are saline tolerant.

133. Answer (3)

Calcium is an immobile element which help in normal functioning of cell membrane and mitotic spindle formation.

134. Answer (1)

Thalassemia is quantitative problem of globin molecules, resulting into anaemia.

135. Answer (4)

Spirogyra shows isogamy.

SECTION - B

136. Answer (2)

Phyllode is modification of leaves.

137. Answer (4)

During flowering photoperiodic stimulus is perceived by leaves of plants.

Vernalisation prevents precocious *i.e.*, early reproductive development.

138. Answer (3)

One Homologus pair means one bivalent. Thus, 24 chromosomes will form 12 bivalents.

139. Answer (4)

$I^AI^B \times I^AI^B$

$\downarrow$

Progeny I^AI^A, I^AI^B, I^AI^B, I^BI^B

Parents cannot have progeny with O blood group.

140. Answer (4)

Big and wrinkled tongue is a feature of individual inflicted with Down's syndrome.

141. Answer (3)

$X^hX \times XY$

$\downarrow$

Progeny X^hX, $\underbrace{X^hY}$, XX, XY

$\downarrow$

Haemophilic progeny (1/4) = 25%

142. Answer (1)

RNA polymerase I synthesises 5.8S, 18S, 28S rRNA.

143. Answer (2)

Prokaryotes have only one type of RNA polymerase but three different DNA polymerases.

144. Answer (3)

ODS (Ozone depleting substances like CFCs) degrade ozone layer and deplete it.

145. Answer (4)

Wildlife sanctuaries are '*in situ*' conservation strategies.

146. Answer (2)

Members of red algae produce non-motile asexual spores and show oogamous sexual reproduction with complex post fertilisation changes.

147. Answer (2)

Mycoplasma are smallest living cells, lack cell wall and can survive without oxygen.

148. Answer (3)

Lac z gene synthesises β-galactosidase.

Lac y gene synthesises permease.

Lac a gene synthesises transacetylase.

i gene synthesises repressor protein.

149. Answer (2)

Down's syndrome is due to autosomal trisomy of chromosome 21.

150. Answer (2)

Phellogen is also called cork cambium which is de-differentiated tissue, *i.e.,* secondary meristematic tissue.

ZOOLOGY

SECTION - A

151. Answer (1)

Chondrichthyes have internal fertilization and many of them are viviparous.

152. Answer (2)

Neurons are the excitable cells of the neural system. The neuroglial cells protect and support neurons.

153. Answer (2)

Cytosine is not a nucleoside, it is a nitrogenous base.

154. Answer (3)

Chylomicrons are transported into the lymph vessels (lacteals) in the villi.

155. Answer (2)

Partial pressure (in mm Hg) of O_2 and CO_2 in different parts is as follows

Respir-atory gas	Atmos-pheric air	Alveoli	Blood deoxy-genated	Blood oxyge-nated	Tissue
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

156. Answer (2)

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

157. Answer (4)

In spiders, excretory structure is coxal gland.

Green gland is present in crustaceans.

158. Answer (3)

Nereis and *Pheretima* are annelids and possess closed circulatory system.

159. Answer (4)

The rods contain a purplish-red protein called the rhodopsin or visual purple, which contains a derivative of vitamin A.

160. Answer (3)

Fallopian tube is lined by simple ciliated columnar epithelium.

161. Answer (4)

For tuberculosis, harmless bacteria vaccine is used.

162. Answer (1)

Uricose glands are present only in male cockroach.

163. Answer (3)

Renal calculi/stone is an insoluble mass of crystallised salts of calcium oxalate formed within the kidneys.

164. Answer (4)

On increasing product concentration, enzymatic activity decreases.

165. Answer (4)

Testosterone is responsible for male secondary sexual characters.

166. Answer (2)

Brush border enzymes are present in intestinal juice.

167. Answer (2)

Monocytes are largest blood cells which are transformed into macrophages.

168. Answer (3)

Vitamin A (Retinol) plays an essential role for growth, vision, differentiation of epithelial tissue.

169. Answer (2)

In actin filament (thin myofilament), cross-arms are absent.

170. Answer (3)

Red muscle fibres have more blood capillaries.

171. Answer (2)

Bowman's gland of nose secrete mucus over the epithelium to keep it moist. Rest of the glands are related to eyes.

172. Answer (3)

Glucagon raises blood glucose level by enhancing the breakdown of glycogen into glucose.

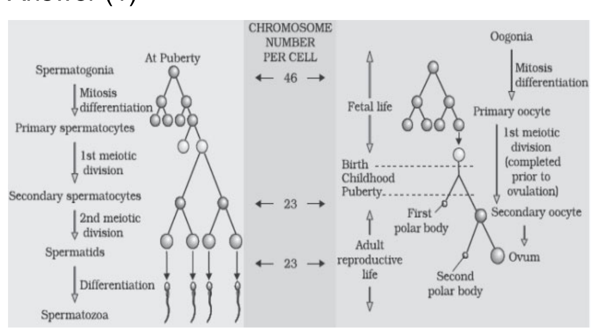
173. Answer (2)

Life expectancy is defined as the age at which half of the population still survives.

174. Answer (1)

300 million × $\frac{60}{100}\times\frac{40}{100}=72$ million sperms per ejaculate

175. Answer (1)



176. Answer (4)
Nirodh is the popular brand for male condom.

177. Answer (3)
Common cold pathogen infects nose, respiratory passage but not lungs.

178. Answer (2)
Small volume cultures (in shake flasks) cannot yield appreciable quantities of product.

179. Answer (4)
High pH leads to association of oxygen and haemoglobin.

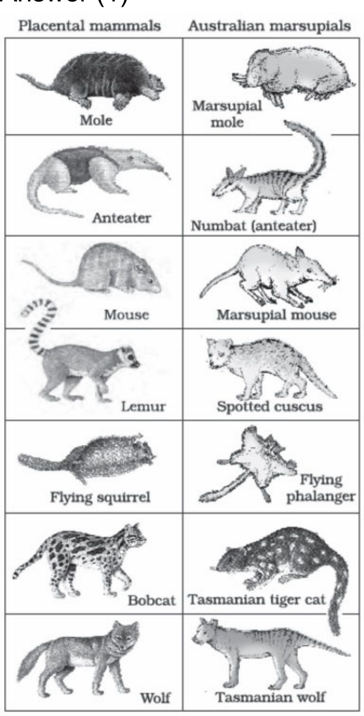
180. Answer (2)
The construction of the 1st rDNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid of *Salmonella typhimurium.*

181. Answer (1)
Basophils release heparin and histamine. Thus, they act like mast cells of connective tissue.

182. Answer (1)
Transgenic plants are ones generated by introducing foreign DNA into a cell and regenerating a plant from that cell.

183. Answer (1)
Maximum water absorption takes place in proximal convoluted tubule.

184. Answer (1)



185. Answer (4)
Structure ‘A’ $\rightarrow$ Pineal gland which is located on the dorsal side of the forebrain.

SECTION - B

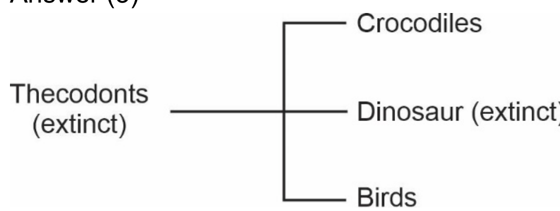
186. Answer (3)
Insulin cannot be orally administered to diabetic patients because it will degrade in alimentary canal.

187. Answer (4)
The 8 bones of the wrist are called carpals. Total number of bones present in cranium is 8.

188. Answer (4)
The zygote or early embryos (with upto 8 blastomeres) can be transferred into the fallopian tube. This technique is known as ZIFT.

189. Answer (1)
Australopithecines probably lived in East African grasslands. Evidence shows that they hunted with stone weapons but essentially ate fruits.

190. Answer (3)



191. Answer (4)
The flower tops, leaves and the resin of *Cannabis* plant are used in various combinations to produce marijuana, hashish, charas and ganja.

192. Answer (1)
New castle disease is the viral disease of poultry.

193. Answer (2)
Catla is a freshwater fish.

194. Answer (3)
Chromosome number in meiocyte of housefly = 12
Chromosome number in gamete of fruit fly = 4

195. Answer (2)
There are an estimated 200,000 varieties of rice in India alone.

196. Answer (4)
High level of progesterone maintains endometrium.

197. Answer (3)
Gill slits are absent in non-chordates.

198. Answer (1)
Phallic gland secretes the outermost layer of spermatophore.

199. Answer (2)
Salivary glands, mammary glands and sweat glands are examples of multicellular glands.

200. Answer (1)
Bipolar neurons ae found in the retina layer of the eye.

❑ ❑ ❑